Learning Networks

Promit K. Chaudhuri [*] Jonghyun Park[†] Sudipta Sarangi[‡]
Hector Tzavellas[§]

January 2026

Abstract

We study learning in network games where agents know only their local neighborhoods. Agents update beliefs about the true network by removing networks inconsistent with observed neighbor actions. We show that learning ends in finite time, and after learning ends, their behavior coincides exactly with the complete-information Nash equilibrium, even if agents may not know the true network. We provide conditions for the perfect learning based on the injectivity of equilibrium actions and show that local symmetries can hinder learning about the true network. Furthermore, we bound the learning speed. Also, under an injective condition, learning ends within a time proportional to the network diameter. Finally, we apply our result to a linear-quadratic model to get a sharp insight.

JEL Classifications: C72, D81, D85

Keywords: Incomplete Information, Network Games, Network Uncertainty, Bayesian Learning

[*]Department of Economics, Virginia Tech, Blacksburg, USA, Email: promit@vt.edu
[†]Department of Economics, Virginia Tech, Blacksburg, USA, Email: jongh2419@vt.edu
[‡]Department of Economics, Virginia Tech, Blacksburg, USA, Email: ssarangi@vt.edu
[§]Department of Economics, Virginia Tech, Blacksburg, USA, Email: hectortz@vt.edu

1 Introduction

1.1 Overview

Network games are a standard tool for modeling environments in which agents interact locally but are affected by the broader architecture of connections. In such settings, equilibrium behavior depends on the *entire* topology: an agent's optimal action is shaped not only by her direct neighbors, but also by the incentives of neighbors-of-neighbors and, more generally, by the global pattern of strategic interdependence.

A tension arises between this theoretical analysis, which assumes that players understand the network structure, and the information limits that agents face in reality. In many cases, agents might not know the whole network. A firm may be aware of its direct suppliers but not the entire supply chain; an individual may know their direct contacts but not beyond that. Empirically, (Breza et al., 2018) shows that agents' knowledge about the network is limited. This gap raises a basic question for applying equilibrium analysis: when agents observe only local neighborhoods and interact with them repeatedly, how can their behavior be shaped, and when do they know the entire structure at the limit?

This paper provides some outcomes under this incomplete environment. Nature draws an (un)directed and (un)weighted network g from a finite set $\mathcal{G}$. Each agent observes only her in-neighbors (incoming links) after the drawing and, in every period, observes only the actions of her in-neighbors. Agents are Bayesian but myopic: in each period, they play the unique Bayesian Nash equilibrium, maximizing the current stage period given their current information, and then update their beliefs about the true network. The key idea is that observing a neighbor's action eliminates some network because the action depends on the neighbor's local information about the network. Thus, actions can be interpreted as signals about the network.

Our main results consider three aspects of learning. First, we show that learning stabilizes in finite time, which naturally raises the following question: What does the equilibrium look like after learning ends? We demonstrate that once learning ends, equilibrium play coincides with the complete-information Nash equilibrium under the realized network. We emphasize the distinction between *knowing the network* and *behaving as if the network were known*: even if agents play the complete Nash equilibrium, they may not know the true network.

Second, we analyze identification: When an agent knows the realized network. We provide sufficient conditions for identification based on the injectivity of complete-information equilibrium actions: if some complete Nash-equilibrium actions of my neighbors or mine under a graph can be distinguished from other graphs' actions, then the agent eventually identifies the realized graph. This idea is related to the symmetry of the equilibrium. When two graphs are isomorphic, it may be difficult to distinguish the equilibrium, which makes identification fail.

Third, we measure the speed of learning. We provide two complementary bounds on the learning time. After showing a general bound that is not restricted to the network structure, we show another bound that depends on the network structure under an assumption. Since each agent observes only in-neighbors, information can transfer to an agent only along directed paths in a network. Under an action-separation condition (distinct types induce distinct equilibrium actions), we can induce a bound that depends on the diameter of the graph.

We apply these results to the canonical model: the linear-quadratic network game of Ballester et al. (2006). In that environment, complete-information equilibrium actions are Katz–Bonacich (KB) centralities, so our equilibrium convergence result suggests a prediction in the long run when players confront incomplete information about the network. Also, we show that the KB centrality can be informative enough to distinguish networks from the equilibrium, which may induce perfect learning about the network. We also analyze how network structure classes, such as a nested split graph, can generate learning advantages for central players through their high connectivity.

1.2 Related Literature

Our analysis is related to social learning and network games.

First, it contributes to the literature on the relationship between equilibrium under incomplete and complete information. Jordan (1995) and Kalai and Lehrer (1993a) show convergence of Bayesian Nash equilibrium to Nash equilibrium in a repeated game. Our framework differs in two dimensions: (i) we consider imperfect monitoring of the equilibrium actions of others, and (ii) instead of showing the convergence to the complete Nash, we show the actions are coincident in a finite time.

Second, the paper contributes to Bayesian learning literature. Because full Bayesian updating is often intractable in large networks, much of the literature studies boundedly rational or non-Bayesian updating rules such as DeGroot learning and related dynamics (Bala and Goyal (1998), Golub and Jackson (2010), Acemoglu et al. (2010)). Our analysis is complementary: we consider full rationality and Bayesian updating by providing the equilibrium prediction, identification, and learning speed.

There is some Bayesian learning literature (Parikh and Krasucki (1990), Mueller-Frank

(2013)). These works emphasize imitation and consensus in the limit by assuming a symmetric payoff setting. Since players have the same payoff function and it may not depend on neighbors' actions, they can imitate a neighbor's action if it yields a higher payoff than their own action. In contrast, we allow general payoff functions with local spillovers and heterogeneity, so convergence need not require the consensus or imitation. Instead, the limiting outcome is that the complete Nash equilibrium is equal to the Bayesian Nash equilibrium.

Third, the paper contributes to the literature on network games with incomplete information. We build on the canonical framework of Ballester et al. (2006), recently extended by Zenou and Zhou (2022) to provide tractable equilibrium characterizations in richer environments. While existing work on incomplete information often assumes the network is known but agents are uncertain about payoff parameters (e.g., Martí and Zenou (2015)), we study the complementary case: payoff parameters are known, but the *network itself* is uncertain.

We also build on the stage-game framework with local observation studied in Chaudhuri et al. (2025), extending it to an infinitely repeated setting to characterize the evolution of beliefs and behavior.

Finally, our results on learning speed connect to work on network structure and dynamic propagation. We show that a symmetry environment can hinder the learning. This perspective complements dynamic contagion models (e.g., Leister et al. (2022)) by providing an explicit Bayesian channel linking observation, belief updating, and convergence of strategic behavior.

Section 2 presents the model, information structure, and updating rule. Section 3 establishes finite-time stabilization and shows that, once learning ends, the Bayesian equilibrium coincides with the complete-information Nash equilibrium; it also provides sufficient conditions for perfect learning. Section 4 shows bounds on learning time. Section 5 applies the framework to linear-quadratic network games. Section 6 concludes. Proofs and some extended results are collected in the Appendix.

2 The Game

This section defines the environment, information structure, and equilibrium concept. The network is fixed over time, but is initially unknown beyond each agent's local incoming neighborhood. Since equilibrium actions are type-contingent, agents can use observed neighbor actions as signals about the underlying network and about what others have learned.

Time and players

Time is discrete and indexed by $t \in \{1, 2, \dots\}$. The finite set of players is $N = \{1, \dots, n\}$.

Networks and ex-ante uncertainty

A network is an $n \times n$ matrix $g = (g_{ji})_{j,i \in N}$, where g_{ji} measures the marginal impact of player j's action on player i's payoff (interpreted as a directed edge from j to i). A network is undirected if $g_{ji} = g_{ij}$ for all $i \neq j$. Let $\mathcal{G}$ be a finite set of feasible networks.[1] Nature draws $g \in \mathcal{G}$ at $t = 1$ according to a common prior $p \in \Delta(\mathcal{G})$, and g remains fixed thereafter. We assume no self-loop: $g_{ii} = 0$ for any $i \in N$.

For each $i \in N$, define the set of in-neighbors

$$N_i^{in}(g) \ := \ \{j \in N : \ g_{ji} \neq 0\}.$$

These are precisely the players whose actions directly enter i's payoff under g.

Information and timing within a period

At $t = 1$, after Nature draws g, each player i observes her incoming link vector $(g_{ji})_{j \in N}$ (equivalently, the ith column of g). Thus, each player knows whom she observes and who directly affects her payoff, but may be uncertain about the remainder of the network.

The within-period timing is:

1. At the start of period t, each player i has an information set I_i^t (defined below) and forms a posterior over $\mathcal{G}$.
2. Players simultaneously choose actions $a_i^t \in A$.
3. Payoffs $u_i(\cdot)$ are realized.
4. Each player i observes the actions $\{a_j^t : \ j \in N_i^{in}(g)\}$ of her in-neighbors, and updates her information set to I_i^{t+1}.

We assume players are *myopic but Bayesian*: in each period they play the Bayesian Nash equilibrium of the current stage game induced by I_i^t, and then update their beliefs using Bayes' rule. Players do not strategically experiment to improve future information.

[1]We consider an infinite $\mathcal{G}$ in the Appendix. If we consider undirected and unweighted network classes, $\mathcal{G}$ is always finite.

Types as partitions of $\mathcal{G}$

Because uncertainty concerns the realized network, it is natural to represent information directly in terms of the set of networks that remain feasible. This representation makes two features of the model transparent. First, a player's *type* is simply a coarse description of the realized network—a subset of $\mathcal{G}$. Second, at each date, the collection of possible types for a given player forms a *partition* of $\mathcal{G}$, so types can be interpreted as the cells of an information partition generated by local observations and observed actions.[2]

Definition 1 (Types as information partitions)**.** Fix a period t. For each $i \in N$ and each $g \in \mathcal{G}$, let $T^t_{i(g)} \subseteq \mathcal{G}$ denote the set of networks that are feasible for player i at the start of period t when the realized network is g.

Define a relation $\sim^t_i$ on $\mathcal{G}$ by

$$g \sim^t_i g' \quad \Longleftrightarrow \quad T^t_{i(g)} = T^t_{i(g')}.$$

Let T^t_i be the collection of equivalence classes (cells) induced by $\sim^t_i$; we refer to each cell $T^t_{i(g)}$ as a *period-t type* of player i.

When Nature draws $g \in \mathcal{G}$, player i's realized type at period t is the unique cell $I^t_i \in T^t_i$ such that $g \in I^t_i$. The realized types $(I^t_i)_{i\in N}$ are private information, while the partitions $(T^t_i)_{i\in N}$ are common knowledge.

Each possible network g generates a well-defined period-t information set for player i, summarized by the feasible set $T^t_{i(g)}$. Group networks together whenever they generate the *same* feasible set for i. This grouping necessarily produces a partition: (i) it is *exhaustive* because every $g \in \mathcal{G}$ generates some feasible set and hence belongs to some group; and (ii) it is *disjoint* because a given network g cannot generate two different feasible sets at the same time, so it cannot belong to two different groups. Thus, the cells of T^t_i split the state space $\mathcal{G}$ into mutually exclusive and collectively exhaustive information classes.

In this framework, learning corresponds to a progressive refinement of these partitions over time: for each i, T^{t+1}_i (weakly) refines T^t_i, because additional observations can only eliminate networks from feasibility and hence can only split existing cells into smaller subsets.

At $t = 1$, since no actions have yet been observed, player i can only condition on her observed incoming links. Hence,

$$T^1_{i(g)} = \Big\{ g' \in \mathcal{G} \ : \ g'_{ji} = g_{ji} \quad \forall j \in N \Big\}. \tag{1}$$

[2]We show the partitioning property mathematically at the end of this section.

When the graph is unweighted and undirected, $N_i^{in}(g)$ coincides with the usual (neighbor) set $N_i(g)$.

Given a realized type $I_i^t = T_{i(g)}^t$, player i's posterior is the prior restricted to the feasible set:

$$p(g' \mid T_{i(g)}^t) = \begin{cases} \dfrac{p(g')}{\sum_{g'' \in T_{i(g)}^t} p(g'')} & \text{if } g' \in T_{i(g)}^t, \\ 0 & \text{otherwise.} \end{cases}$$

The denominator is the total probability mass assigned to i's feasible set, and the numerator is the probability mass of the particular network g'.

The equilibrium actions (which will be defined below) are functions of types. As a result, observing an action allows players to rule out networks under which the relevant type would have generated a different equilibrium action.

Payoffs, local spillovers, and the stage game

Player i's payoff under network g depends on her own action, the actions of her in-neighbors, and (possibly) directly on the local architecture:

$$u_i\Big(a_i,\ (a_j)_{j \in N_i^{in}(g)},\ g\Big).$$

Unlike much of the observational learning literature, we do not impose identical preferences across agents. In standard settings with common objectives and no payoff externalities, an agent can often replicate a neighbor's action and thus cannot perform worse on average.[3] Once preferences differ or actions create externalities, copying need not be optimal, and learning must proceed through the structure of equilibrium actions rather than through imitation.

We consider a special type of network game. A given player's payoff depends on other players' actions, but only on those to whom the player is linked in the network. In fact, without loss of generality, the network can be taken to indicate the payoff interactions in the society (Jackson and Zenou (2015)). Following this idea, we impose a standard locality restriction: remote edges can affect i only through their impact on neighbors' actions, not directly in u_i.

Assumption 1 (Local Spillovers)**.** *Player i's payoff depends on the network only through*

[3]The imitation principle; see Gale and Kariv (2003) and Mueller-Frank (2013)

her incoming neighborhood. Formally, for any $g, g' \in \mathcal{G}$,

$$u_i\Big(a_i, (a_j)_{j \in N_i^{in}(g)}, g\Big) = u_i\Big(a_i, (a_j)_{j \in N_i^{in}(g')}, g'\Big) \quad \textit{whenever } g'_{ji} = g_{ji}\ \forall j \in N.$$

Assumption 1 implies that if two networks agree on i's incoming links (and their weights), then they are payoff-equivalent from i's perspective in that period. This is the sense in which spillovers are *local*: remote parts of the network affect i only indirectly through how they shape others' equilibrium actions.

This is the structural component common to the class of anonymous interaction games discussed in Galeotti et al. (2010) (Section 3). In those settings, payoffs are further restricted to be *anonymous* within the neighborhood, depending on neighbors' actions only through an aggregate statistic (e.g., the sum or average), or more generally being invariant to relabeling of neighbors. Our assumption fixes the scope of direct interaction—that only connected peers matter—while remaining general enough to allow for identity-dependent weights (as in linear-quadratic models) or other non-anonymous heterogeneity.

The following example shows the widely used payoff function in network game contexts that satisfies local spillovers.

Example 1 (Linear-quadratic network game (Ballester et al. (2006)))**.** *Let $A = \mathbb{R}$ and*

$$u_i\Big(a_i, (a_j)_{j \in N_i^{in}(g)}, g\Big) = a_i - \frac{1}{2}a_i^2 + \lambda a_i \sum_{j=1}^{n} g_{ji} a_j,$$

where λ measures the strength of local complementarities. (In the unweighted undirected case, $g_{ji} \in \{0, 1\}$ and $g_{ji} = g_{ij}$.)

Fix a period t and a (common-knowledge) strategy profile $\sigma^t = (\sigma_i^t)_{i \in N}$, where $\sigma_i^t : T_i^t \to A$ maps types to actions. Given $I_i^t = T_{i(g)}^t$, player i evaluates current expected payoff as

$$\mathbb{E}\big[u_i \mid I_i^t = T_{i(g)}^t\big] = \sum_{g' \in \mathcal{G}} p(g' \mid T_{i(g)}^t)\, u_i\Big(a_i,\ (a_j)_{j \in N_i^{in}(g')},\ g'\Big), \tag{2}$$

where, under the counterfactual network g', each opponent j plays the action prescribed by σ_j^t at the corresponding type $T_{j(g')}^t$.

Definition 2 (Stage-t Bayesian Nash equilibrium)**.** A pure strategy profile σ^{*t} is a Bayesian Nash equilibrium (BNE) of the period-t stage game if for every $i \in N$ and every $T_{i(g)}^t \in T_i^t$,

$$\sigma_i^{*t}(T_{i(g)}^t) \in \arg\max_{a_i \in A}\ \mathbb{E}\big[u_i(a_i, \sigma_{-i}^{*t}) \mid I_i^t = T_{i(g)}^t\big]\,.$$

Since our goal is to focus on the learning process rather than on equilibrium characterization, we assume the existence and uniqueness of the pure BNE at each period.

Assumption 2 (Existence and uniqueness)**.** *For each t, the stage game admits a unique pure BNE.*

Assumption 2 plays two roles. First, it makes equilibrium behavior a single-valued (measurable) function of agents' time-t information, so observed actions can be used as well-defined signals for inference about others' types and about the network. Second, it rules out equilibrium-selection effects that would otherwise be confounded with learning: when a stage game admits multiple equilibria, long-run play can depend on priors or conventions that pick among equilibria rather than on the informational content of observed behavior.

In applications, uniqueness is guaranteed under standard sufficient conditions. For example, in linear-quadratic network games, uniqueness obtains under the mild spectral-radius restrictions that make the best-response operator a contraction (Ballester et al. (2006), Bramoullé et al. (2014)).

If the stage game admits multiple pure BNEs, all arguments go through under the following weaker requirement: there exists a measurable equilibrium selection rule that assigns to each period-t type profile a particular pure BNE, and this selection rule is common knowledge (Kreps, 1990; Young, 1993). In that case, we interpret $a^t(\cdot)$ as the selected equilibrium action profile.

Dynamic updating from observed actions

At the end of period t, player i observes $(a_j^t)_{j \in N_i^{in}(g)}$. Because equilibrium strategies are type-contingent, an observed action restricts the set of neighbor types (and hence networks) that could have generated it.

Definition 3. For a neighbor j and an observed action value $a_j \in A$ at time t, define the set of j-types consistent with a_j by

$$B^t(a_j) := \left\{ T_{j(g')}^t \in T_j^t \ : \ \sigma_j^{*t}(T_{j(g')}^t) = a_j \right\}.$$

Let the corresponding set of graphs consistent with observing $\bar{a}$ be

$$B_f^t(a_j) := \bigcup_{S \in B^t(a_j)} S.$$

Thus, $B^t(a_j)$ collects all neighbor types that would induce the observed action $\bar{a}$ in equilibrium.[4] Since a type is a set of feasible networks, $B_f^t(a_j)$ is the set of networks that remain feasible *from j's perspective* given the observed action. Player i updates by intersecting her previous feasible set with the graphs that are simultaneously consistent with all neighbor actions she observes.

Definition 4 (Type updating rule)**.** For each $i \in N$, $g \in \mathcal{G}$, define

$$T_{i(g)}^t = \begin{cases} \{g' \in \mathcal{G} : g'_{ji} = g_{ji} \ \forall j \in N\} & t = 1, \\ T_{i(g)}^{t-1} \cap \Big(\bigcap_{j \in N_i^{in}(g)} B_f^{t-1}(a_j^{t-1}) \Big) & t \geq 2. \end{cases}$$

The updating rule is well-defined because all graphs in $T_{i(g)}^{t-1}$ share the same incoming neighborhood of i (so i knows which neighbors' actions are observed).

Basic properties

We record several properties that will be used repeatedly.

Remark 1 (Nonemptiness)**.** *For all $i \in N$, $g \in \mathcal{G}$, and $t \geq 1$, we have $g \in T_{i(g)}^t$. In particular, $T_{i(g)}^t \neq \emptyset$.*

This implies the type set is non-empty.

Remark 2 (Common in-neighborhood within a type)**.** *For any $i \in N$ and any $t \geq 1$, if $g, g' \in T_{i(g)}^t$, then $g'_{ji} = g_{ji}$ for all $j \in N$, and hence $N_i^{in}(g) = N_i^{in}(g')$.*

This shows that if two graphs are in the same type set for i, their local information should be the same.

The lemma combining with the remark 1 below shows that the type set shows a partition of $\mathcal{G}$ by showing each type set is mutually exclusive.

Lemma 1 (Types form a partition of $\mathcal{G}$)**.** *For any $i \in N$ and $t \geq 1$, and any $g, g' \in \mathcal{G}$, either $T_{i(g)}^t = T_{i(g')}^t$ or $T_{i(g)}^t \cap T_{i(g')}^t = \emptyset$.*

Finally, we show a property about the type set. At each period, by observing the neighbors' actions, players eliminate some graphs, which implies a type set is non-decreasing.[5]

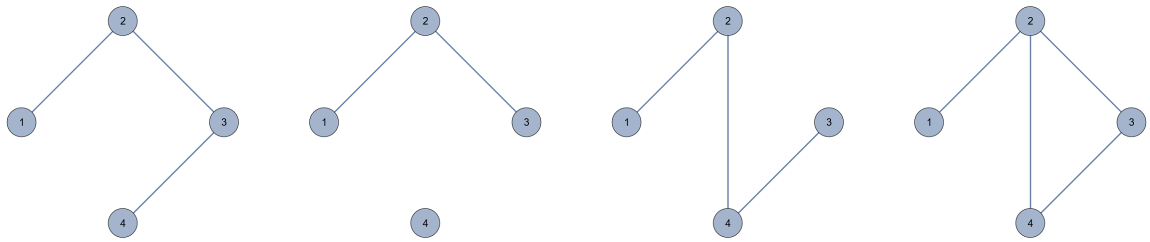
Remark 3 (Monotonicity and finiteness)**.** *For each i and g, the sequence $(T_{i(g)}^t)_{t \geq 1}$ is non-increasing in t. Since $\mathcal{G}$ is finite, each $T_{i(g)}^t$ is finite and the updating process can refine $T_{i(g)}^t$ only finitely many times.*

[4] The set $B^t(a_j)$ may contain multiple types if different types induce the same equilibrium action.

[5] This can be directly proved from Definition 4.

2.1 Example of the Learning Process

We illustrate the definitions in a four-player environment (Figure 1) where $\mathcal{G} = \{g_a, g_b, g_c, g_d\}$ and payoffs are linear-quadratic as in Example 1.[6] Assume priors $(0.1, 0.2, 0.3, 0.4)$ over (g_a, g_b, g_c, g_d) and suppose Nature selects g_a.



(a) g_a (b) g_b (c) g_c (d) g_d

Figure 1: A four-player example with $\mathcal{G} = \{g_a, g_b, g_c, g_d\}$.

Stage 1 (direct inference from a neighbor). Under g_a realization, player 2 observes an initial type $I_2^1 = T^1_{2(g_a)} = \{g_a, g_b\}$, because under g_c player 2 has a different neighborhood than under g_a.[7] Player 1 observes $I_1^1 = T^1_{1(g_a)} = \{g_a, g_b, g_c, g_d\}$ since player 1's neighborhood is identical across all candidate graphs.

In the stage-1 BNE, suppose player 2's equilibrium action differs across her possible types:

$$a_2^1(T^1_{2(g_a)}) = a_2^1(T^1_{2(g_b)}) = 1.789, \qquad a_2^1(T^1_{2(g_c)}) = 1.877, \qquad a_2^1(T^1_{2(g_d)}) = 2.387.$$

Player 1 observes $a_2^1(T^1_{2(g_a)}) = 1.789$ and therefore infers that player 2's realized type must be $T^1_{2(g_a)} = T^1_{2(g_b)} = \{g_a, g_b\}$. Hence, player 1 updates to

$$I_1^2 = T^2_{1(g_a)} = T^1_{1(g_a)} \cap B_f^1(a_2^1) = \{g_a, g_b\}.$$

Stage 2 (higher-order learning). At $t = 2$, player 1 knows $g \in \{g_a, g_b\}$. Under either graph, player 2 observes player 3's stage-1 action, which differs across player 3's possible types,[8] and thus player 2 perfectly learns whether $g = g_a$ or $g = g_b$ by the end of stage 1. Anticipating this, player 1 expects player 2's stage-2 action to depend on which graph is

[6] In this illustration, we assume the graphs are unweighted and undirected, so $N_i^{in}(g)$ coincides with the usual neighbor set $N_i(g)$.

[7] For instance, $N_2(g_a) = N_2(g_b) = \{1, 3\}$ but $N_2(g_c) = \{1, 4\}$.

[8] Player 2 observes player 3's action, and $a_3^1(I_3^1 = T^1_{3(g_a)}) = 2.041 \neq a_3^1(I_3^1 = T^1_{3(g_b)}) = 1.447$. Thus, player 2 can identify the true graph after observing player 3's action.

realized. If, in the stage-2 BNE,

$$a_2^2(T^2_{2(g_a)}) = 1.813 \quad \text{and} \quad a_2^2(T^2_{2(g_b)}) = 1.716,$$

then observing $a_2^2 = 1.813$ allows player 1 to deduce $g = g_a$. This illustrates how local observation can generate higher-order inference: by watching a neighbor, a player can learn about what the neighbor learned from others.

3 Convergence and Identification

This section shows agents' behavior and their information about the true network after learning ends. Two notions should be kept conceptually separate.

First, agents may eventually *behave as if* they know the true network: the BNE coincides with the complete-information Nash equilibrium. However, this does not imply agents know the realized structure. Second, agents may (or may not) *identify* the realized network. Our results show that the equilibrium coincidence is guaranteed in a finite time, while the identification depends on whether equilibrium behavior provides a sufficiently rich "action signature" to break observational equivalences (and may fail under symmetries).

3.1 A bridge to complete information

A key feature of our environment is that learning is *local*: agents observe only their neighbors' actions, not beyond this direct information. We show that under this environment, the BNE is equal to the complete Nash in the long run. This subsection demonstrates the paper's main result: once learning ends, the Bayesian Bash equilibrium coincides *exactly* with the complete-information Nash equilibrium.

We begin by formalizing when the learning process stops changing agents' information sets. Recall from Section 2 that for each realized network $g \in \mathcal{G}$ and player i, the set $T^t_{i(g)} \subseteq \mathcal{G}$ is the collection of networks that remain feasible for player i at the start of period t.

Definition 5 (Learning ends)**.** Learning ends at stage t^* if for all $t \geq t^*$,

$$T^t_{i(g)} = T^{t^*}_{i(g)} \qquad \text{for all } i \in N \text{ and } g \in \mathcal{G}.$$

Definition 5 captures the content of learning in this environment: after t^*, no agent rules out any additional candidate networks based on observed behavior. Importantly, this does *not* require that agents have already pinned down the true network; it only requires that the equilibrium path ceases to generate new eliminations.

The first result is immediate but foundational: since feasible sets are finite and shrink only by elimination, the updating process cannot refine information forever.

Lemma 2 (Finite-time termination)**.** *There exists a finite t^* such that learning ends.*

Lemma 2 follows from the monotonicity of updating: for each (i, g), the sequence $(T^t_{i(g)})_{t\geq 1}$ is (weakly) decreasing and takes values in the finite set $2^{\mathcal{G}}$, so it can change only finitely many times.

We now state the main result of the section. It shows that stabilization is not merely an informational stopping time—it is a *behavioral* stopping time. Once learning ends, any residual network uncertainty becomes observationally and payoff-wise irrelevant on the equilibrium path, so play matches the complete-information benchmark exactly.

Let $a^c(g)$ denote the unique pure Nash equilibrium action profile of the one-shot stage game when the realized network g is common knowledge, and let $a^c_i(g)$ be player i's component.

Theorem 1. *Suppose Assumptions 1–2 hold and Nature selects $g \in \mathcal{G}$. If learning ends at t^*, then for all $t \geq t^*$,*

$$a^t_i\big(I^t_i = T^t_{i(g)}\big) = a^c_i(g) \qquad \forall i \in N.$$

Theorem 1 is the paper's core behavioral implication. Agents may stop short of identifying the realized topology—indeed, $|T^{t^*}_{i(g)}|$ may exceed one—yet they still behave *as if* the entire network were common knowledge. Therefore, structural uncertainty does not impede strategic efficiency in the long run.

Stabilization forces a strong form of observational equivalence on each agent's observed margin: every surviving graphs makes the same predictions about what the agent will actually see. Under local spillovers, those predictions are exactly what pins down best responses, so the incomplete-information equilibrium collapses to the complete-information fixed point.

While some seminal works (Kalai and Lehrer (1993a) and Jordan (1995)) guarantee that Bayesian learning leads to Nash equilibrium *asymptotically*—often implying convergence only in distribution or to an ϵ-neighborhood—Theorem 1 departs from their classical settings in two critical ways. First, we replace global monitoring with strictly *local* observation—agents see only their neighbors' actions. Second, we strengthen the convergence result: rather than asymptotic convergence, we prove a *finite-time bridge* where the incomplete-information equilibrium coincides *exactly* with the complete-information benchmark after a bounded number of periods. We show the upper bound for the learning time in Section 4.

3.2 Identification and perfect learning

Theorem 1 shows that agents eventually behave as if the network were known. We now ask when agents actually *learn* the network itself.

Definition 6 (Perfect learning)**.** Learning is perfect for player i if, at the end of learning, i's information set is a singleton:

$$T_{i(g)}^{t^*} = \{g\}.$$

Perfect learning requires that equilibrium behavior be sufficiently discriminating to rule out every incorrect network, not merely to pin down equilibrium actions. A simple sufficient condition is that some observed player's complete-information action uniquely encodes the network.

Theorem 2. *Suppose Assumptions 1–2 hold, $g \in \mathcal{G}$ is realized, and learning ends at t^*. Fix $i \in N$. If either of the following conditions holds, then player i perfectly learns the true network:*

- *There exists a neighbor $j \in N_i^{in}(g)$ such that the mapping $g' \mapsto a_j^c(g')$ is one-to-one on $\mathcal{G}$.*
- *The mapping $g' \mapsto a_i^c(g')$ is one-to-one on $\mathcal{G}$.*

Theorem 2 highlights the role of equilibrium structure in identification. If an agent observes (or can infer from her own behavior) an action whose complete-information level is unique to the realized network, then stabilization forces the feasible set to collapse to the true singleton $\{g\}$. In many parametric classes, this injectivity is generic, but it can fail systematically when candidate networks admit symmetries that preserve what an agent can observe.[9]

The next result formalizes a symmetry-based obstruction. It shows that even when behavior converges to the complete-information benchmark, isomorphic alternatives may remain feasible forever because they generate identical local observational implications.

Proposition 1. *Let $\mathcal{G}^n$ be the set of all undirected and unweighted networks on N with a uniform prior distribution p. Let $g \in \mathcal{G}^n$ be realized and learning ends at t^*. If there exists an isomorphic network $g' \neq g$ and a bijection $f : N \to N$ satisfying $f(i) = i$ and $f(j) = j$ for all $j \in N_i^{in}(g)$,*[10] *then*

$$g' \in T_{i(g)}^{t^*}.$$

Proposition 1 clarifies the conceptual distinction emphasized throughout the paper. Behavioral convergence to the complete-information Nash equilibrium is robust: it requires only

[9] We revisit this in the linear-quadratic application in Section 5.

[10] For undirected networks, $N_i^{in}(g)$ coincides with the usual neighbor set of i.

that the remaining candidate networks be observationally equivalent along the equilibrium path. Perfect identification is stricter and may fail when the environment admits symmetries that cannot be broken under local monitoring.

The results in this section are qualitative: they establish finite-time convergence to the complete-information benchmark and characterize when the realized network is (or is not) identified. The next section turns to a quantitative question: how long can the learning phase last, and how does that duration depend on $|\mathcal{G}|$ and on network topology?

4 Learning Speed

Sections 2–3 establish two qualitative facts: (i) agents' feasible sets shrink only by elimination and therefore stabilize in finite time, and (ii) once learning stabilizes, equilibrium play coincides with the complete-information Nash equilibrium. This section asks a quantitative question: *how long can the elimination process last?*

Two distinct constraints discipline the speed of learning.

1. *A combinatorial constraint.* Since each type is a subset of the finite set $\mathcal{G}$, a player's information can be refined only finitely many times. This yields a coarse bound that depends only on n and $|\mathcal{G}|$.

2. *A propagation constraint.* Under local monitoring, information about the realized network can reach an agent only through neighbors' actions, and in directed networks, only along directed paths that end at the observer. Under an action-separation condition that will be defined below, this yields a sharper bound in terms of directed distances.

4.1 A coarse bound from finite hypothesis elimination

We begin with a bound that does not require any injectivity or separation of equilibrium actions. The key object is the partition of $\mathcal{G}$ induced by each player's types. For each player i and period t, define the collection of i's type sets by

$$\mathcal{T}_i^t \ := \ \{\, T_{i(g)}^t \ : \ g \in \mathcal{G} \,\}, \qquad b_i^t \ := \ |\mathcal{T}_i^t|.$$

Learning progresses precisely when at least one player's partition becomes strictly finer. Since each partition is a partition of a finite set, it can strictly refine only finitely many times.

Theorem 3 (Coarse uniform bound)**.** *Suppose Assumptions 1–2 hold. Let t^* be the time at which learning ends. Then*

$$t^* \;\leq\; 1 + n\big(|\mathcal{G}| - 1\big).$$

Theorem 3 does not require that observed actions uniquely identify types (i.e., actions are injective into types), and it allows for pooling, symmetry equilibrium actions, and other sources of slow elimination. The bound corresponds to the worst case in which, in each period prior to termination, only one player refines her feasible set and does so by eliminating only one additional graph in $\mathcal{G}$.

4.2 A directed distance bound under action separation

The coarse bound above uses only the finiteness of $\mathcal{G}$. To bring topology into the bound, we need that observed actions transmit sufficiently sharp information about neighbors' information. If different types can induce the same equilibrium action, then observing an action may fail to reveal what a neighbor has learned, and higher-order inference may stall even in small-diameter networks. We therefore impose an action-separation condition used only in this subsection.[11]

Assumption 3 (Action separation)**.** *For each player i and each period t, distinct types induce distinct equilibrium actions: if $T^t_{i(g)} \neq T^t_{i(g')}$, then*

$$a^t_i\big(T^t_{i(g)}\big) \neq a^t_i\big(T^t_{i(g')}\big)\,,$$

where $a^t_i(\cdot)$ denotes player i's (unique) equilibrium action at time t as a function of her period-t type.

Assumption 3 has a clean informational meaning: by observing a^t_i, an observer can infer player i's period-t type exactly. This makes it possible for information to propagate iteratively: by watching a neighbor, a player can learn not only about the neighbor's local links (known at $t = 1$) but also about what the neighbor learned from others in previous periods.[12]

The assumption is naturally satisfied in many continuous-action environments. In particular, in the linear-quadratic class studied later, equilibrium actions depend smoothly (indeed, analytically) on network statistics, so distinct informational states generically produce distinct actions except under structural symmetries.

[11] Also, Assumption 3 automatically implies the sufficient conditions for Theorem 2. Therefore, under Assumptions 1-3, the agents can learn perfectly.

[12] For the sufficient conditions or related discussion for this assumption, see Mueller-Frank (2013) and Arieli and Mueller-Frank (2017).

Because the model allows directed links and observation is of *in-neighbors*, information can reach player i only from nodes that are upstream of i along directed paths that end at i. We formalize this with directed distance.

For a directed network g, define the directed distance from j to i by

$$dist_g(j,i) \ := \ \min\Big\{\ell \in \mathbb{N} : \ \exists \text{ a directed path } j \to \cdots \to i \text{ of length } \ell\Big\},$$

with the convention $dist_g(j,i) = \infty$ if no such path exists. For $r \geq 0$, define the upstream r-neighborhood of i by

$$N_i^r(g) \ := \ \{\, j \in N : \ dist_g(j,i) \leq r \,\}.$$

Thus, $N_i^1(g) = N_i^{in}(g) \cup \{i\}$, and $N_i^r(g)$ expands by adding the in-neighbors of the current set. The quantity $dist_g(j,i)$ measures how many rounds of "who learned what from whom" must elapse before information originating at j can, in principle, be reflected in what i observes.

Define the directed diameter of g by

$$diam(g) \ := \ \max_{(i,j)\in N} \{\, dist_g(j,i) : \ dist_g(j,i) < \infty \,\},$$

and the maximal directed diameter across candidates by

$$d(\mathcal{G}) \ := \ \max_{g \in \mathcal{G}} diam(g).$$

Proposition 2. *Suppose Assumptions 1–3 hold. Then learning ends by stage*

$$t^* \ \leq \ d(\mathcal{G}) + 1.$$

Proposition 2 captures a simple propagation logic. At $t = 1$, each player j's type $T^1_{j(g)}$ is pinned down by j's links. Under action separation, observing a^1_j reveals $T^1_{j(g)}$ exactly. Therefore, by the end of the first period, each player i can infer the realized incoming neighborhoods of her in-neighbors, i.e., she effectively learns the portion of the network within directed distance 2 upstream. Iterating this argument, after r rounds a player can infer what her upstream nodes at directed distance r must have known at $t = 1$, and therefore no genuinely new eliminations remain once r reaches the largest finite directed distance in the candidate class. When $r \geq d(\mathcal{G})$, all information that could ever reach any player under local monitoring has already done so, and feasible sets stabilize.

Furthermore, since $diam(g) \leq n-1$ for any graph g with n-players, we can know under Assumptions 1–3, the learning time is linear in the number of players.

Corollary 1. *Under Assumptions 1–3,*

$$t^* \leq n.$$

The bounds in this section are complementary. Theorem 3 requires only finiteness and uniqueness and applies even when actions pool across types; Proposition 2 is sharper but relies on action separation to ensure that observed behavior fully reveals neighbors' types. In Section 5 we apply the general framework to a canonical parametric class (the linear-quadratic network game) to illustrate when separation and identification are plausible and how topology shapes speed and informativeness.

5 Application: Linear-Quadratic Network Games

This section illustrates the general learning framework in a canonical parametric class: the linear-quadratic network game of Ballester et al. (2006). Throughout this section, we specialize to *unweighted, undirected* networks for transparency. Formally, each network $g \in \mathcal{G}$ is an adjacency matrix $G = (g_{ij})_{i,j \in N}$ with $g_{ij} \in \{0,1\}$, $g_{ij} = g_{ji}$ for all $i \neq j$, and $g_{ii} = 0$. We interpret each undirected edge as a bilateral payoff interaction and a bilateral observation link. Hence, under local monitoring, player i observes the actions of her (undirected) neighbors $N_i(g) := \{j : \ g_{ij} = 1\}$, and the directed-distance balls in Section 4 reduce to the usual graph-distance neighborhoods.

The application serves two purposes. First, it provides a closed-form complete-information benchmark $a^c(g)$, making Theorem 1 especially transparent: once learning ends, play must coincide with a centrality-based action profile. Second, the closed form yields concrete conditions under which equilibrium actions separate networks (facilitating identification), and it clarifies how symmetry can block perfect learning.

5.1 The linear-quadratic environment

Fix $A = \mathbb{R}$ and let payoffs be

$$u_i(a, g) \;=\; a_i - \frac{1}{2}a_i^2 + \lambda\, a_i \sum_{j=1}^{n} g_{ij} a_j, \qquad i \in N, \tag{3}$$

where $\lambda > 0$ measures the strength of local complementarities and $g_{ij} \in \{0, 1\}$ indicates whether i and j are linked. The game satisfies the *local spillovers* property in Assumption 1: player i's payoff depends on the network only through her neighborhood $N_i(g)$ and the actions of those neighbors.

The unique pure Nash equilibrium of this complete-information game is

$$a^c(g) \;=\; (I - \lambda G)^{-1}\mathbf{1}, \tag{4}$$

whenever $I - \lambda G$ is invertible. A sufficient condition is $0 < \lambda < 1/\rho(G)$, where $\rho(G)$ is the spectral radius of G. Expression (4) is the Katz–Bonacich centrality vector of the undirected network with parameter λ. Thus, under complete information, equilibrium behavior is centrality-based: actions increase with an agent's total exposure to others through walks of all lengths.

We now connect the application back to the general results. Let $g \in \mathcal{G}$ be the realized network and let t^* be the time at which learning ends (Definition 5). By Theorem 1, once learning ends, the repeated-game equilibrium actions must match the complete-information equilibrium actions under the true g. Combining this with (4) yields an immediate implication.

Corollary 2. *Suppose Assumptions 1–2 hold and payoffs are given by* (3)*. If learning ends at t^* under realized g, then for all $t \geq t^*$,*

$$a_i^t\big(I_i^t = T_{i(g)}^t\big) = \big[(I - \lambda G)^{-1}\mathbf{1}\big]_i \qquad \forall i \in N.$$

Even though agents initially observe only their own local neighborhood and infer the rest of the network indirectly through observed actions, the eventual action profile is identical to the one that would arise if the entire network were common knowledge. Residual uncertainty (if any) can survive only over networks that are behaviorally indistinguishable on the equilibrium path.

5.2 Identification via network walks

The centrality representation also clarifies when actions can identify g. Using the Neumann series expansion, for $0 < \lambda < 1/\rho(G)$,

$$a^c(g) = (I - \lambda G)^{-1}\mathbf{1} = \sum_{k=0}^{\infty} \lambda^k G^k \mathbf{1}. \tag{5}$$

For each $k \geq 0$, define the k-step *walk count* at i by

$$W_i^k(g) \;:=\; (G^k \mathbf{1})_i.$$

Because G is the 0–1 adjacency matrix of an undirected network, $W_i^k(g)$ is the number of length-k walks that end at i (counting multiplicities). Equation (5) implies

$$a_i^c(g) = \sum_{k=0}^{\infty} \lambda^k W_i^k(g),$$

so $a_i^c(g)$ is a generating function of i's walk counts. In particular, two networks generate different complete-information actions for player i as soon as they differ in walk counts at some length.

Proposition 3. *Fix $i \in N$ and two networks $g, g' \in \mathcal{G}$. Suppose there exists $k \geq 1$ such that $W_i^k(g) \neq W_i^k(g')$. Then the function*

$$\lambda \mapsto a_i^c(g) - a_i^c(g')$$

is not identically zero on any interval where both equilibria are well-defined. In particular, for all λ except a (possibly empty) finite set,

$$a_i^c(g) \neq a_i^c(g').$$

The intuition is immediate from (5). Equilibrium actions aggregate the entire walk structure of the network around the player: if two networks differ in the number of walks reaching i at some length, then the resulting generating functions cannot coincide for more than finitely many values of λ. Since agents observe actions, differences in walk counts translate into observable differences, enabling identification.

5.3 A symmetry obstruction: local isomorphism

We now make precise how symmetry can block perfect learning. The key idea is that if two networks differ only by relabeling nodes in parts of the graph that are never anchored by player i's observable region, then i may be unable to distinguish them from any finite history of observed neighbor actions.

Fix a realized network g and player i. Let $N_i^r(g)$ denote the radius-r neighborhood of i under *undirected* graph distance:

$$N_i^r(g) = \{j \in N : \ \mathrm{dist}_g(i,j) \leq r\}.$$

Definition 7 (Local isomorphism around i)**.** Let $g, g' \in \mathcal{G}$ be two networks and fix $i \in N$. We say that g' is *locally isomorphic to* g *around* i *at radius* r if there exists a bijection $f : N \to N$ such that:

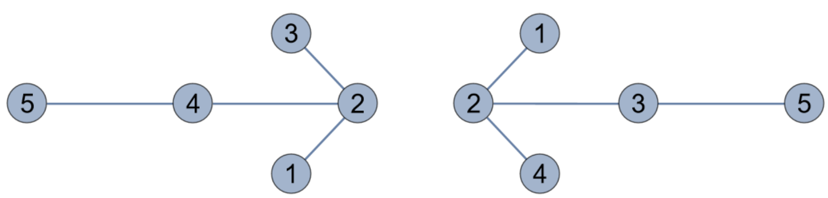
1. $f(j) = j$ for all $j \in N_i^r(g)$ (the radius-r neighborhood is fixed), and
2. for all $u, v \in N$, the adjacency indicators satisfy $g_{uv} = g'_{f(u)f(v)}$ (graph isomorphism under relabeling).

Local isomorphism means that g' is obtained from g by permuting only nodes *outside* a specified neighborhood of i, while preserving the overall network structure. If the permutation fixes a sufficiently large neighborhood, then the entire collection of walk counts around i is preserved, and so are the equilibrium actions observed by i.

Proposition 4. *Consider the linear-quadratic game* (3)*. If there exists $g' \neq g$ such that for every radius $r \geq 1$ the network g' is locally isomorphic to g around i at radius r (Definition 7), then i cannot perfectly learn: for any terminal time t^* at which learning ends,*

$$g' \in T_{i(g)}^{t^*}.$$

Proposition 4 formalizes the distinction between *behavioral convergence* and *identification.* Even when play converges to $a^c(g)$, perfect learning can fail if the hypothesis class contains networks that are observationally indistinguishable from i's vantage point.



(a) g_a (b) g_b

Figure 2: $\mathcal{G}^{iso} = \{g_a, g_b\}$. The networks are isomorphic under a permutation that swaps nodes 3 and 4.

Consider the two graphs g_a and g_b in the figure, which are isomorphic via the bijection f that swaps nodes 3 and 4 (i.e., $f(3) = 4$ and $f(4) = 3$) while fixing nodes 1, 2, and 5. Let g_a be realized. Player 1's initial type is $I_1^1 = \{g_a, g_b\}$. Player 1's only neighbor is 2, and the isomorphism fixes node 2. Consequently, player 1 observes identical neighbor behavior under g_a and g_b and cannot distinguish them.

In contrast, player 2 observes nodes 3 and 4 directly. Because nodes 3 and 4 are swapped by the isomorphism but occupy different positions relative to node 5, their equilibrium actions differ across the two networks. Hence, player 2 distinguishes the graphs quickly.

5.4 Accelerated learning in nested split graph classes

Finally, we consider how central positions facilitate faster learning within a structured hypothesis class. We focus on *nested split graphs* (also called threshold graphs), which are split graphs in which periphery neighborhoods are totally ordered by inclusion (König et al., 2014). This class accommodates *multiple hierarchy levels*: different periphery nodes can be connected to progressively larger subsets of the clique, yielding several degree tiers (star networks are an extreme special case).

Let $\mathcal{G}^{nest} \subseteq \mathcal{G}$ be a hypothesis class of nested split graphs on N. Each $g \in \mathcal{G}^{nest}$ admits a partition $N = C \cup P$ such that: (i) C is a clique, (ii) P is an independent set, and (iii) periphery neighborhoods into the clique are nested: there exists an ordering $p_1, \dots, p_m$ of P with

$$N_{p_1}(g) \cap C \ \subseteq \ N_{p_2}(g) \cap C \ \subseteq \ \cdots \ \subseteq \ N_{p_m}(g) \cap C, \tag{6}$$

where $N_p(g) = \{j \in N : \ g_{pj} = 1\}$.

Condition (6) implies a clear hierarchy: a higher-tier periphery node is linked to a superset of the clique neighbors of any lower-tier periphery node. In particular, if every $p \in P$ has at least one clique neighbor (no isolated periphery), then the smallest set $N_{p_1}(g) \cap C$ is nonempty, and any clique vertex in this set is adjacent to *every* periphery node. Such vertices have maximal degree in the realized graph.

We call a player *core* if she is a maximal-degree clique vertex (equivalently, a clique vertex adjacent to all periphery nodes, when the periphery has no isolated vertices). Intuitively, core players have the widest field of view: they observe all clique actions (since C is a clique) and, in addition, they observe the entire periphery.

We also rule out degenerate cases where some player is already fully informed at $t = 1$.

Definition 8 (No initial information advantage)**.** We say the game has no initial information advantage if for every $i \in N$ and every realized $g \in \mathcal{G}$, the initial feasible set is not already terminal: $T_{i(g)}^{t^*} \subsetneq T_{i(g)}^{1}$ whenever learning ends at t^*.

This condition rules out trivial cases where the network is identified immediately at $t = 1$, ensuring that the learning dynamics are non-degenerate.

Proposition 5. *Consider the linear-quadratic game (3). Suppose $\mathcal{G} = \mathcal{G}^{nest}$, the periphery has no isolated vertices, and the game has no initial information advantage (Definition 8). Then:*

1. *Core players achieve perfect learning for all λ except a possibly empty finite set.*
2. *If, in addition, the equilibrium action function is injective in types along the equilibrium path, then core players' learning times are weakly shorter than those of periphery players.*

Nested split graphs impose a visibility hierarchy: core players observe many actions each period, and the nesting property ensures that these observations separate candidate graphs quickly. In the linear-quadratic model, actions encode walk structure; core players effectively see enough of the periphery to break many symmetries early. Proposition 5 therefore formalizes a learning advantage of centrality within a structured hypothesis class: core players are not only more influential (as captured by Katz–Bonacich centrality), but also better positioned to infer the realized network under local monitoring.

6 Concluding Remarks

This paper bridges the theoretical gap between standard network games, which assume complete structural knowledge, and the empirical reality where agents possess only local vision. Our central contribution is demonstrating that this disconnect is transient: in environments with local spillovers, structural uncertainty is eventually rendered behaviorally irrelevant. We show that rational agents, by observing the actions of their neighbors, effectively decode the network structure until their equilibrium behavior coincides exactly with the Nash Equilibrium of the complete-information game. This result provides a microfoundation for the robustness of complete-information models, justifying their use as long-run predictors even in opaque environments.

After showing the convergence result, we provide conditions for perfect learning. Also, we consider the speed of learning and how a network structure (a diameter) can bound it under an assumption. We applied this framework to the canonical linear-quadratic network game, illustrating that agents eventually converge to actions characterized by their Katz-Bonacich centrality.

Future research may extend our framework to environments with non-myopic agents who strategically experiment to accelerate information learning.

Appendix A: Proofs

Proof of Lemma 1

Fix $i \in N$ and $t \geq 1$. We show that the family $\{T^t_{i(g)}\}_{g\in\mathcal{G}}$ is a partition of $\mathcal{G}$: for any $g, g' \in \mathcal{G}$, either $T^t_{i(g)} = T^t_{i(g')}$ or $T^t_{i(g)} \cap T^t_{i(g')} = \emptyset$.

Step 1 (Base case $t = 1$). Recall that

$$T^1_{i(g)} = \left\{ h \in \mathcal{G} : \; h_{ji} = g_{ji} \text{ for all } j \in N \right\}.$$

Suppose $T^1_{i(g)} \cap T^1_{i(g')} \neq \emptyset$, and pick h in the intersection. Then for every $j \in N$,

$$g_{ji} = h_{ji} = g'_{ji}.$$

Hence, g and g' have the same incoming-link vector into i, which implies $T^1_{i(g)} = T^1_{i(g')}$ by the defining equality above. Therefore, at $t = 1$ the sets are either identical or disjoint.

Step 2 (Induction step). Fix $t \geq 1$ and assume the claim holds at time t, i.e., for any $g, g' \in \mathcal{G}$, either $T^t_{i(g)} = T^t_{i(g')}$ or $T^t_{i(g)} \cap T^t_{i(g')} = \emptyset$. We prove it holds at $t + 1$.

If $T^t_{i(g)} \cap T^t_{i(g')} = \emptyset$, then since types are updated by intersection, $T^{t+1}_{i(g)} \subseteq T^t_{i(g)}$ and $T^{t+1}_{i(g')} \subseteq T^t_{i(g')}$, so $T^{t+1}_{i(g)} \cap T^{t+1}_{i(g')} = \emptyset$ as well.

Now suppose $T^t_{i(g)} = T^t_{i(g')}$. Denote this common set by S. We show $T^{t+1}_{i(g)} = T^{t+1}_{i(g')}$. By the updating rule, for each realized network $\tilde{g}$,

$$T^{t+1}_{i(\tilde{g})} = T^t_{i(\tilde{g})} \; \cap \bigcap_{j \in N^{in}_i(\tilde{g})} B^t_f\Big(a^t_j\big(I^t_j = T^t_{j(\tilde{g})}\big)\Big).$$

Since $T^t_{i(g)} = T^t_{i(g')}$, player i has the same information at time t under g and under g'. In particular, i's in-neighborhood is the same across all networks in S (otherwise the sets in Step 1 would already be disjoint). Hence the index set of observed neighbors is the same under g and g':

$$N^{in}_i(g) = N^{in}_i(g').$$

Moreover, for each such neighbor j, the observed equilibrium action at time t is the same under g and g' because the common information set S induces the same period-t type for j:

$$a^t_j\big(I^t_j = T^t_{j(g)}\big) = a^t_j\big(I^t_j = T^t_{j(g')}\big).$$

Therefore the same family of sets is intersected with S under g and under g', which implies

$T^{t+1}_{i(g)} = T^{t+1}_{i(g')}$.

Combining the two cases proves that $\{T^{t+1}_{i(g)}\}_{g\in\mathcal{G}}$ also consists of pairwise identical-or-disjoint sets. Hence $\{T^{t}_{i(g)}\}_{g\in\mathcal{G}}$ is a partition of $\mathcal{G}$ for every t.

■

Proof of Lemma 2

Fix (i, g). By construction, $g \in T^t_{i(g)}$ for all t, so $T^t_{i(g)}$ is always nonempty. By the updating rule,

$$T^{t+1}_{i(g)} = T^t_{i(g)} \;\cap \bigcap_{j\in N^{in}_i(g)} B^t_f\Big(a^t_j\big(I^t_j = T^t_{j(g)}\big)\Big),$$

so $T^{t+1}_{i(g)} \subseteq T^t_{i(g)}$ for all t: the sequence is weakly decreasing under set inclusion.

Since $\mathcal{G}$ is finite, there cannot exist an infinite strictly decreasing chain of nonempty subsets of $\mathcal{G}$. Therefore, for each (i, g) there exists $t_{i,g}$ such that $T^t_{i(g)} = T^{t_{i,g}}_{i(g)}$ for all $t \geq t_{i,g}$. Let $t^* := \max_{i\in N,\ g\in\mathcal{G}} t_{i,g}$. Then for all $t \geq t^*$,

$$T^t_{i(g)} = T^{t^*}_{i(g)} \qquad \text{for all } i \in N,\ g \in \mathcal{G},$$

which means learning ends.

■

Proof of Theorem 1

Fix a realized network $g \in \mathcal{G}$ and suppose learning ends at t^*. Fix any $t \geq t^*$.

Step 0 (Payoff-relevant neighborhood is known). By the construction of initial types in Section 2, player i knows her incoming neighborhood under the realized network at $t = 1$, and updating only eliminates networks from feasibility. Hence, for every t and every $g' \in T^t_{i(g)}$,

$$N^{in}_i(g') = N^{in}_i(g). \tag{7}$$

In particular, conditional on $I^t_i = T^t_{i(g)}$, player i assigns positive probability only to networks that share the same in-neighbor set as g.

Step 1 (Observational equivalence after stabilization). We record the key implication of learning ending.

Claim 1. *For any $i \in N$, any $t \geq t^*$, and any $g', g'' \in T^t_{i(g)}$,*

$$a^t_j\big(I^t_j = T^t_{j(g')}\big) = a^t_j\big(I^t_j = T^t_{j(g'')}\big) \qquad \forall\, j \in N^{in}_i(g).$$

Proof of Claim 1 Fix i, $t \geq t^*$, and $g', g'' \in T^t_{i(g)}$. Take any observed in-neighbor $j \in N^{in}_i(g)$. Suppose, towards a contradiction, that

$$a^t_j\big(I^t_j = T^t_{j(g')}\big) \neq a^t_j\big(I^t_j = T^t_{j(g'')}\big)\,.$$

Under the realized network g, player i observes j's period-t action $a^t_j\Big(I^t_j = T^t_{j(g)}\Big)$. Since g' and g'' predict different period-t actions for j, at most one of them can be consistent with the observed action. Therefore, updating must eliminate at least one of $\{g', g''\}$ from $T^t_{i(g)}$, contradicting that learning has ended at $t^* \leq t$.

■

Step 2 (Conditional payoffs collapse to complete-information payoffs). Fix player i. Consider her expected period-t payoff from choosing an action a_i, conditional on her type $I^t_i = T^t_{i(g)}$. Let $p(\cdot \mid T^t_{i(g)})$ denote her posterior over the feasible set $T^t_{i(g)}$. Under Assumption 1 (local spillovers) and (7), the payoff under any candidate $g' \in T^t_{i(g)}$ depends on g' only through the in-neighbor set (which is fixed and equal to $N^{in}_i(g)$) and the induced in-neighbor actions. Thus,

$$\mathbb{E}\big[u_i \mid I^t_i = T^t_{i(g)}\big] = \sum_{g' \in T^t_{i(g)}} p(g' \mid T^t_{i(g)})\, u_i\Big(a_i,\ \big(a^t_j(I^t_j = T^t_{j(g')})\big)_{j \in N^{in}_i(g)},\ g\Big). \tag{8}$$

By Claim 1, for each $j \in N^{in}_i(g)$ the term $a^t_j(I^t_j = T^t_{j(g')})$ is the same for all $g' \in T^t_{i(g)}$ and equals the realized neighbor action $a^t_j(I^t_j = T^t_{j(g)})$. Therefore the sum in (8) collapses to

$$\mathbb{E}\big[u_i \mid I^t_i = T^t_{i(g)}\big] = u_i\Big(a_i,\ \big(a^t_j(I^t_j = T^t_{j(g)})\big)_{j \in N^{in}_i(g)},\ g\Big). \tag{9}$$

Step 3 (The stabilized profile is the complete-information equilibrium). Because $\big(a^t_i(I^t_i)\big)_{i \in N}$ is a Bayesian Nash equilibrium at time t, for each i the chosen action $a^t_i(I^t_i = T^t_{i(g)})$ maximizes the left-hand side of (9) over a_i. By (9), it therefore maximizes the right-hand side of (9). Hence, for each i, $a^t_i(I^t_i = T^t_{i(g)})$ is a best response in the *complete-information* stage game under g to the in-neighbor action vector $\big(a^t_j(I^t_j = T^t_{j(g)})\big)_{j \in N^{in}_i(g)}$.

Since this argument holds for every $i \in N$, the action profile $\big(a^t_i(I^t_i = T^t_{i(g)})\big)_{i \in N}$ is a pure Nash equilibrium of the complete-information stage game under g. By Assumption 2 (uniqueness of the pure Nash equilibrium under complete information), this profile must equal $a^c(g)$. Therefore,

$$a^t_i\big(I^t_i = T^t_{i(g)}\big) = a^c_i(g) \qquad \forall\, i \in N.$$

Because $t \geq t^*$ was arbitrary, the conclusion holds for all $t \geq t^*$, completing the proof.

■

Proof of Theorem 2

Let learning end at t^* and fix player i and realized network g.

Case 1 (observed injective neighbor). Suppose there exists $j \in N_i^{in}(g)$ such that the mapping $g' \mapsto a_j^c(g')$ is injective on $\mathcal{G}$. Suppose, toward a contradiction, that player i does not learn perfectly, so there exists $g' \neq g$ with $g' \in T_{i(g)}^{t^*}$.

By Claim 1 applied at time t^*, for the observed neighbor j we have

$$a_j^{t^*}\left(I_j^{t^*} = T_{j(g)}^{t^*}\right) = a_j^{t^*}\left(I_j^{t^*} = T_{j(g')}^{t^*}\right).$$

By Theorem 1, for any network h, $a_j^{t^*}(I_j^{t^*} = T_{j(h)}^{t^*}) = a_j^c(h)$. Hence the equality above implies $a_j^c(g) = a_j^c(g')$, contradicting injectivity.

Case 2 (own-action injectivity). Suppose instead that the mapping $g' \mapsto a_i^c(g')$ is injective on $\mathcal{G}$. If i does not learn perfectly, there exists $g' \neq g$ with $g' \in T_{i(g)}^{t^*}$. Since $g' \in T_{i(g)}^{t^*}$ implies $T_{i(g')}^{t^*} = T_{i(g)}^{t^*}$, we have

$$a_i^{t^*}\left(I_i^{t^*} = T_{i(g)}^{t^*}\right) = a_i^{t^*}\left(I_i^{t^*} = T_{i(g')}^{t^*}\right).$$

Applying Theorem 1 yields $a_i^c(g) = a_i^c(g')$, contradicting injectivity. Thus in either case, $T_{i(g)}^{t^*} = \{g\}$ and i learns perfectly.

■

Proof of Proposition 1

Let $g, g' \in \mathcal{G}^n$ be isomorphic, and let $f : N \to N$ be an isomorphism such that $f(i) = i$ and $f(j) = j$ for all $j \in N_i^{in}(g)$. Since f fixes i and all of i's observed neighbors, g and g' have the same incoming links into i, and hence $g' \in T_{i(g)}^1$.

Because the prior is uniform over $\mathcal{G}^n$ and the stage-game equilibrium is unique, the equilibrium correspondence is equivariant under relabeling: if $a^t(\cdot)$ denotes the period-t equilibrium action profile as a function of types, then applying f to relabel the network and the players produces the relabeled action profile. In particular, for every period t and every neighbor $j \in N_i^{in}(g)$,

$$a_j^t\big(I_j^t = T_{j(g)}^t\big) = a_{f(j)}^t\big(I_{f(j)}^t = T_{f(j)(g')}^t\big) = a_j^t\big(I_j^t = T_{j(g')}^t\big),$$

where the last equality uses $f(j) = j$.

Thus, along the equilibrium path, player i observes exactly the same sequence of neighbor actions under g and under g'. Therefore i never eliminates g' from her feasible set, and $g' \in T^{t^*}_{i(g)}$ when learning ends at t^*.

■

Proof of Theorem 3

Fix $t \geq 1$ and recall that for each player i, $\mathcal{T}_i^t = \{T^t_{i(g)} : g \in \mathcal{G}\}$ is a partition of $\mathcal{G}$ (Lemma 1). Let $b_i^t := |\mathcal{T}_i^t|$ be the number of cells in i's partition at time t, and define the aggregate count

$$B^t := \sum_{i \in N} b_i^t.$$

Since each $\mathcal{T}_i^t$ partitions $\mathcal{G}$, we have $1 \leq b_i^t \leq |\mathcal{G}|$ and hence

$$n \leq B^t \leq n|\mathcal{G}| \qquad \text{for all } t.$$

Step 1. If learning has not ended at period t, then $B^{t+1} \geq B^t + 1$.

Indeed, suppose no player refines at t, i.e., $T^{t+1}_{i(g)} = T^t_{i(g)}$ for all i and g. Then every player faces the same type at $t+1$ as at t, and since the period-t equilibrium is unique, the equilibrium action profile at $t+1$ coincides with that at t. Consequently, every player observes exactly the same neighbor actions at $t+1$ as at t, and the same reasoning applies inductively: no further eliminations can ever occur. Therefore, learning must already have ended at t. The contrapositive implies that if learning has not ended at t, at least one player strictly refines, hence for some i, $b_i^{t+1} \geq b_i^t + 1$, and thus $B^{t+1} \geq B^t + 1$.

Step 2. The total number of strict refinements is bounded by $n(|\mathcal{G}| - 1)$.

Since B^t can increase by at least 1 in each period in which learning is active (Step 1), and since $B^t \leq n|\mathcal{G}|$ always while $B^1 \geq n$, the number of periods in which B^t can strictly increase is at most $n|\mathcal{G}| - n = n(|\mathcal{G}| - 1)$. As the process starts at $t = 1$, it follows that learning ends by

$$t^* \leq 1 + n(|\mathcal{G}| - 1),$$

which proves the theorem.

■

Proof of Proposition 2. Fix a realized network g and a player i.

Step 0 (Define the object that can be learned). For $r \geq 1$, define the *upstream path subgraph* of depth r for player i by

$$H_i^r(g) \;:=\; \big(N_i^r(g),\, E_i^r(g)\big),$$

where $N_i^r(g) := \{j : dist_g(j,i) \le r\}$ (as in the main text) and

$$E_i^r(g) \;:=\; \{(k \to \ell) \in E(g) :\; dist_g(\ell, i) \le r-1\}.$$

Equivalently, $E_i^r(g)$ consists exactly of those edges that lie on some directed path of length $\le r$ terminating at i. This is the maximal portion of g that can ever affect what i observes through the direction of information flow.

Step 1 (Induction claim, correctly indexed). We claim: for every $t \ge 1$, at the *start of period* t player i can infer $H_i^t(g)$.

Base case ($t = 1$). At the start of period 1, player i observes her incoming-link vector $(g_{ji})_{j \in N}$ and the corresponding weights (if any). Hence she knows $N_i^1(g) = \{i\} \cup N_i^{in}(g)$ and she knows all edges in $E_i^1(g)$ (these are exactly the edges with head i). Therefore she can infer $H_i^1(g)$.

Inductive step. Assume the claim holds for some $t \ge 1$. Consider the start of period $t+1$. At the end of period t, player i observes a_j^t for each $j \in N_i^{in}(g)$. By Assumption 3 (action separation), observing a_j^t pins down the realized type $T_{j(g)}^t$ for each such neighbor j.

Now apply the induction hypothesis *to player* j: at the start of period t, player j can infer $H_j^t(g)$; equivalently, all networks in $T_{j(g)}^t$ agree on $H_j^t(g)$. Since i has identified $T_{j(g)}^t$, she can therefore infer $H_j^t(g)$ for every observed neighbor $j \in N_i^{in}(g)$.

Finally, note that any node k with $dist_g(k,i) \le t+1$ either already lies in $N_i^t(g)$ or there exists a directed path $k \to \cdots \to j \to i$ with $j \in N_i^{in}(g)$ and $dist_g(k,j) \le t$. Thus $k \in N_j^t(g)$ for some observed neighbor j, and every edge on a directed path of length $\le t+1$ into i is either an edge into i (known already) or an edge on a directed path of length $\le t$ into some such j (contained in $H_j^t(g)$, hence inferred by i). Therefore i can infer $H_i^{t+1}(g)$.

This completes the induction.

Step 2 (Conclude the learning-time bound). Let $d(\mathcal{G}) = \max_{g' \in \mathcal{G}} diam(g')$ as defined in the main text. For the realized g, any directed path that can ever transmit information into any player i has length at most $diam(g) \le d(\mathcal{G})$. Hence by the induction claim, at the start of period $d(\mathcal{G}) + 1$ every player has inferred the entire upstream path subgraph that could possibly affect what she can observe in the future. Consequently, no further observation can refine any player's feasible set, so learning must have ended by stage $t^* \le d(\mathcal{G}) + 1$.

■

Proof of Corollary 2

Fix a realized network $g \in \mathcal{G}$ and suppose learning ends at t^*. By Theorem 1, for every $t \geq t^*$ the period-t equilibrium action profile coincides with the unique complete-information Nash equilibrium under g, i.e., $a_i^t(I_i^t = T_{i(g)}^t) = a_i^c(g)$ for all i.

Under the linear-quadratic payoff (3), the unique complete-information equilibrium (whenever it exists) is given by (4), namely $a^c(g) = (I - \lambda G)^{-1}\mathbf{1}$. Substituting this expression yields the claim.

■

Proof of Proposition 3 Fix $i \in N$ and two networks $g, g' \in \mathcal{G}$ with adjacency matrices G, G'. Define

$$f(\lambda) \;:=\; a_i^c(g) - a_i^c(g') \;=\; e_i^\top (I - \lambda G)^{-1}\mathbf{1} \;-\; e_i^\top (I - \lambda G')^{-1}\mathbf{1},$$

where e_i is the ith standard basis vector.

Step 1 (Neumann expansion and non-identicality). For any λ such that both inverses exist (e.g., for $\lambda \in \big(0,\, 1/\max\{\rho(G), \rho(G')\}\big)$), the Neumann series yields

$$(I - \lambda G)^{-1}\mathbf{1} = \sum_{k=0}^{\infty} \lambda^k G^k \mathbf{1}, \qquad (I - \lambda G')^{-1}\mathbf{1} = \sum_{k=0}^{\infty} \lambda^k (G')^k \mathbf{1}.$$

Hence

$$f(\lambda) = \sum_{k=0}^{\infty} \lambda^k \Big((G^k \mathbf{1})_i - ((G')^k \mathbf{1})_i \Big) = \sum_{k=0}^{\infty} \lambda^k \Big(W_i^k(g) - W_i^k(g') \Big).$$

If $W_i^k(g) \neq W_i^k(g')$ for some $k \geq 1$, then the coefficient on λ^k is nonzero, so $f(\lambda)$ cannot be the zero function on any interval where the expansion is valid.

Step 2 (Finitely many exceptional λ). Each coordinate of $(I - \lambda G)^{-1}\mathbf{1}$ is a rational function of λ:

$$e_i^\top (I - \lambda G)^{-1}\mathbf{1} = \frac{P_i(\lambda)}{\det(I - \lambda G)}$$

for some polynomial $P_i(\lambda)$, and similarly for G'. Therefore, $f(\lambda) = N_i(\lambda)/D(\lambda)$ where N_i is a polynomial and $D(\lambda) \neq 0$ whenever both inverses exist. Since f is not identically zero (Step 1), N_i is not the zero polynomial, so it has only finitely many roots. Hence, $f(\lambda) = 0$ for at most finitely many λ in any interval where both equilibria are well-defined, and in particular $a_i^c(g) \neq a_i^c(g')$ except possibly for a finite set of λ.

■

Proof of Proposition 4

Fix realized g and player i, and suppose there exists $g' \neq g$ such that for every $r \geq 1$ the network g' is locally isomorphic to g around i at radius r (Definition 7).

We prove that $g' \in T^t_{i(g)}$ for all t, hence in particular $g' \in T^{t^*}_{i(g)}$ at any terminal time t^* when learning ends.

Step 1 (A finite-speed observation lemma). Let $H^t_i(g)$ denote player i's *observable history* up to period t under network g, i.e., the sequence consisting of i's initially observed neighborhood together with the time-$1, \ldots, t-1$ actions of her neighbors. Under local monitoring, information can propagate at most one edge per period, so $H^t_i(g)$ depends only on the induced subgraph on the radius-t neighborhood $N^t_i(g)$. Formally, one proves by induction on t that the neighbors' actions observed by i at time s are measurable with respect to the induced subgraph on $N^{s+1}_i(g)$, and therefore the entire history $H^t_i(g)$ is measurable with respect to the induced subgraph on $N^t_i(g)$.

Step 2 (Local isomorphism implies identical observable history). Fix any $t \geq 1$ and take $r = t$. By assumption, there exists a bijection $f : N \to N$ such that $f(j) = j$ for all $j \in N^t_i(g)$ and $g_{uv} = g'_{f(u)f(v)}$ for all $u, v \in N$. In particular, the subgraphs induced by the vertex set $N^t_i(g)$ coincide. Let $g\restriction_S$ denote the induced subgraph on set S (containing all edges (u, v) such that $u, v \in S$). Then:

$$g\restriction_{N^t_i(g)} = g'\restriction_{N^t_i(g)} .$$

Because payoffs (3), monitoring, and the equilibrium selection rule are invariant under relabeling (e.g., under a uniform prior, or more generally under a prior invariant to graph isomorphisms), Step 1 implies

$$H^t_i(g) = H^t_i(g').$$

Step 3 (No elimination). Player i updates her feasible set by eliminating graphs inconsistent with her observable history. Since $H^t_i(g) = H^t_i(g')$ for every t, the graph g' is never ruled out. Equivalently, $g' \in T^t_{i(g)}$ for all t, and in particular $g' \in T^{t^*}_{i(g)}$ at any terminal time t^* when learning ends. Thus, i cannot perfectly learn.

■

Proof of Proposition 5

We provide a proof under the standard definition of nested split graphs and the local-monitoring structure used in the paper.

Part (i): Core players achieve perfect learning. Fix a core player $i \in C$ of maximal degree. Consider any alternative network $g' \in T^1_{i(g)}$ consistent with i's initially observed links. Since $g' \neq g$ and both lie in $\mathcal{G}^{nest}$, there exists at least one periphery node $p \in P$ whose neighborhood in the core differs between g and g':

$$N_g(p) \cap C \;\neq\; N_{g'}(p) \cap C.$$

Because i is a maximal-degree core vertex, i is adjacent to every periphery node, so $p \in N_i(g)$. Therefore, i observes p's equilibrium actions each period.

Under complete information, player p's equilibrium action is Katz–Bonacich centrality $a^c_p(\cdot)$. The two graphs g and g' differ in the walk structure rooted at p, so by Proposition 3, for all λ except a finite set we have $a^c_p(g) \neq a^c_p(g')$. By Corollary 2, once learning ends at t^*, observed actions coincide with complete-information actions. Since i observes p and $a^c_p(g) \neq a^c_p(g')$, the alternative g' is inconsistent with the observed long-run action of p and must be eliminated. Hence $T^{t^*}_{i(g)} = \{g\}$.

Part (ii): Core learning times are weakly shorter under action injectivity. Assume that the equilibrium action function is injective in types along the equilibrium path (the separation condition): for each period t and each player j, the mapping $T \mapsto a^t_j(I^t_j = T)$ is one-to-one on the set of types that arise on the equilibrium path. Fix a realized graph $g \in \mathcal{G}^{nest}$ and let i be a core player.

By the definition of core in the text (maximal-degree clique vertex adjacent to every periphery node), player i is adjacent to every other player. Hence i observes a^1_j for all $j \neq i$ at the end of period $t = 1$.

At $t = 1$, each player j's type $T^1_{j(g)}$ is determined by her observed local neighborhood: in the undirected unweighted specialization, $T^1_{j(g)}$ is exactly the set of networks in $\mathcal{G}$ that coincide with g on the jth row/column of the adjacency matrix (equivalently, that induce the same neighborhood $N_j(g)$). Therefore, knowing $T^1_{j(g)}$ is equivalent to knowing j's neighbor set $N_j(g)$.

By injectivity, player i can infer each neighbor's period-1 type from the observed action: for every $j \neq i$, observing a^1_j uniquely pins down $T^1_{j(g)}$, and therefore pins down $N_j(g)$. In addition, i knows $N_i(g)$ directly at $t = 1$ from her own observation.

Consequently, by the start of period $t = 2$ player i knows the entire collection of neighborhoods $\{N_j(g)\}_{j \in N}$. Since the network is undirected and unweighted, this collection uniquely determines the adjacency matrix: for any unordered pair $\{j, k\}$ we have $g_{jk} = 1$ if and only if $k \in N_j(g)$. Hence, player i reconstructs the entire realized network g by the start of period 2, so her feasible set collapses to $\{g\}$ by $t = 2$.

Now consider any periphery player $p \in P$. By definition of a split graph, P is independent, so p is not adjacent to other periphery nodes and does not observe their period-1 actions. Thus, at the start of period 2, p can infer $N_j(g)$ only for j in her observed neighborhood $\{p\} \cup N_p(g)$, but not for all $j \in N$. Therefore, p cannot generally reconstruct the full adjacency matrix by $t = 2$. It follows that the learning time of core player i is weakly shorter than that of any periphery player.

■

Appendix B: Asymptotic Learning

We consider the case where learning does not necessarily end in finite time. Let $(\mathcal{G}, \mathcal{F})$ be a measurable space, where $\mathcal{G}$ is a set of all possible graphs and $\mathcal{F}$ is a $\sigma-$algebra of subset of $\mathcal{G}$. We assume $\mathcal{F}$ to be generated by the join of partitions. Let p be a common probability measure on $\mathcal{F}$. The other learning structures remain the same to our set-up.

We define $T^{\infty}_{i(g)}$ as

$$T^{\infty}_{i(g)} = \cap_{t=1}^{\infty} T^{t}_{i(g)}.$$

The limit type set $T^{\infty}_{i(g)}$ consists of the possible graphs that are contained in the information set $T^{t}_{i(g)}$ in every period t.

Let $a^{\infty}_i(I^{\infty}_i = T^{\infty}_{i(g)}) = \lim_{t\to\infty} a^t_i(I^t_i = T^t_{i(g)})$.

Then, we can get the following analogous result.

Proposition 6. *Suppose that Nature has selected g. Then, $a^{\infty}_i(I^{\infty}_i = T^{\infty}_{i(g)}) = a^c_i(g), \forall i \in N$ and $\forall g \in \mathcal{G}$, where $a^c_i(g)$ be NE of player i in a complete game under the graph g.*

Proof of Proposition 6

Consider the expected utility at the limit:

$$E[u_i|I^{\infty}_i = T^{\infty}_{i(g)}] = \int_{g'\in\mathcal{G}} p(g'|T^{\infty}_{i(g)}) u_i(a_i(I^{\infty}_i = T^{\infty}_{i(g)}), (a_j(I^{\infty}_j = T^{\infty}_{j(g')}))_{j\in N^{in}_i(g')}, g')$$

Consider a BNE profile $(a^{\infty}_i(I^{\infty}_i = T^{\infty}_{i(g)}))_{i\in N, g\in\mathcal{G}}$. By Assumption 1, we can rewrite

$$\begin{aligned} E[u_i|I^{\infty}_i = T^{\infty}_{i(g)}] &= \int_{g'\in\mathcal{G}} p(g'|T^{\infty}_{i(g)}) u_i(a_i(I^{\infty}_i = T^{\infty}_{i(g)}), (a_j(I^{t}_j = T^{\infty}_{j(g')}))_{j\in N^{in}_i(g')}, g') \\ &= \int_{g'\in\mathcal{G}} p(g'|T^{\infty}_{i(g)}) u_i(a_i(I^{\infty}_i = T^{\infty}_{i(g)}), (a_j(I^{t}_j = T^{\infty}_{j(g')}))_{j\in N^{in}_i(g)}, g). \end{aligned}$$

By Theorem 1, we can rewrite

$$\begin{aligned} &\int_{g'\in\mathcal{G}} p(g'|T^{\infty}_{i(g)}) u_i(a_i(I^{\infty}_i = T^{\infty}_{i(g)}), (a_j(I^{t}_j = T^{\infty}_{j(g')}))_{j\in N^{in}_i(g)}, g) \\ &= \int_{g'\in\mathcal{G}} p(g'|T^{\infty}_{i(g)}) u_i(a_i(I^{\infty}_i = T^{\infty}_{i(g)}), (a_j(I^{\infty}_j = T^{\infty}_{j(g)}))_{j\in N^{in}_i(g)}, g) \\ &= u_i(a_i(I^{\infty}_i = T^{\infty}_{i(g)}), (a_j(I^{\infty}_j = T^{\infty}_{j(g)}))_{j\in N_i(g)}, g) \sum_{g'\in\mathcal{G}} p(g'|T^{\infty}_{i(g)}) \\ &= u_i(a_i(I^{\infty}_i = T^{t}_{i(g)}), (a_j(I^{\infty}_j = T^{\infty}_{j(g)}))_{j\in N_i(g)}, g), \end{aligned}$$

implying that $(a_i(I_i^{\infty} = T^t_{i(g)}))_{i \in N}$ is a NE of a complete network game under the graph g.

From the second to the third line, we use Claim 1 in the proof of Theorem 1, which shows the same action values under graphs that has a positive probability under $T^{\infty}_{i(g)}$.

■

References

Acemoglu, D., A. Ozdaglar, and A. ParandehGheibi (2010). "Spread of (mis)information in social networks". *Games and Economic Behavior* 70.2, pp. 194–227.

Arieli, I. and M. Mueller-Frank (2017). "Inferring beliefs from actions". *Games and Economic Behavior* 102, pp. 455–461.

Bala, V. and S. Goyal (1998). "Learning from neighbours". *The Review of Economic Studies* 65.3, pp. 595–621.

Ballester, C., A. Calvó-Armengol, and Y. Zenou (2006). "Who's who in networks. Wanted: The key player". *Econometrica* 74.5, pp. 1403–1417.

Battigalli, P., F. Panebianco, and P. Pin (2022). "Learning and self-confirming equilibria in network games". *Journal of Economic Theory* 203, p. 105470.

Bramoullé, Y., R. Kranton, and M. D'amours (2014). "Strategic interaction and networks". *American Economic Review* 104.3, pp. 898–930.

Breza, E., A. G. Chandrasekhar, and A. Tahbaz-Salehi (2018). *Seeing the forest for the trees? An investigation of network knowledge*. Tech. rep. National Bureau of Economic Research.

Chaudhuri, P. K. et al. (2025). "Games Under Network Uncertainty". *Working Paper*.

Fudenberg, D. and D. K. Levine (1993). "Steady state learning and Nash equilibrium". *Econometrica: Journal of the Econometric Society*, pp. 547–573.

Gale, D. and S. Kariv (2003). "Bayesian learning in social networks". *Games and economic behavior* 45.2, pp. 329–346.

Galeotti, A. et al. (2010). "Network games". *The review of economic studies* 77.1, pp. 218–244.

Geanakoplos, J. D. and H. M. Polemarchakis (1982). "We can't disagree forever". *Journal of Economic Theory* 28.1, pp. 192–200.

Golub, B. and M. O. Jackson (2010). "Naive learning in social networks and the wisdom of crowds". *American Economic Journal: Microeconomics* 2.1, pp. 112–149.

Huang, W., P. Strack, and O. Tamuz (2024). "Learning in repeated interactions on networks". *Econometrica* 92.1, pp. 1–27.

Jackson, M. O. and Y. Zenou (2015). "Games on networks". In: *Handbook of game theory with economic applications*. Vol. 4. Elsevier, pp. 95–163.

Jordan, J. S. (1991). "Bayesian learning in normal form games". *Games and Economic Behavior* 3.1, pp. 60–81.

Jordan, J. S. (1995). "Bayesian learning in repeated games". *Games and Economic Behavior* 9.1, pp. 8–20.

Kalai, E. and E. Lehrer (1993a). "Rational learning leads to Nash equilibrium". *Econometrica: Journal of the Econometric Society*, pp. 1019–1045.

Kalai, E. and E. Lehrer (1993b). “Subjective equilibrium in repeated games”. *Econometrica: journal of the Econometric Society*, pp. 1231–1240.

Kalai, E. and E. Lehrer (1995). “Subjective equilibrium in repeated games”. *Econometrica* 63.1, pp. 123–143.

König, M. D., C. J. Tessone, and Y. Zenou (2014). “Nestedness in networks: A theoretical model and some applications”. *Theoretical Economics* 9.3, pp. 695–752.

Kreps, D. M. (1990). *Corporate culture and economic theory*. Cambridge University Press.

Leister, M., Y. Zenou, and J. Zhou (2022). “Social connectedness and local contagion”. *The Review of Economic Studies* 89.1, pp. 372–410.

Li, J. and X. Tan (2020). “Locally Bayesian learning in networks”. *Theoretical Economics* 15.1, pp. 239–278.

Mahadev, N. V. and U. N. Peled (1995). *Threshold graphs and related topics*. Vol. 56. Elsevier.

Martí, J. de and Y. Zenou (2015). “Network games with incomplete information”. *Journal of Mathematical Economics* 61, pp. 221–240.

Martí, J. de and Y. Zenou (2016). “Key players under incomplete information”. *Economics Letters* 149, pp. 1–6.

Molavi, P., A. Tahbaz-Salehi, and A. Jadbabaie (2018). “A theory of non-Bayesian social learning”. *Econometrica* 86.2, pp. 445–490.

Mueller-Frank, M. (2013). “A general framework for rational learning in social networks”. *Theoretical Economics* 8.1, pp. 1–40.

Mueller-Frank, M. and C. Neri (2021). “A general analysis of boundedly rational learning in social networks”. *Theoretical Economics* 16.1, pp. 317–357.

Parikh, R. and P. Krasucki (1990). “Communication, consensus, and knowledge”. *Journal of Economic Theory* 52.1, pp. 178–189.

Sadler, E. (2020). “Diffusion games”. *American Economic Review* 110.1, pp. 225–270.

Sun, B., W. Zhao, and J. Zhou (2023). “Structural interventions in networks”. *International Economic Review* 64.4, pp. 1557–1589.

Young, H. P. (1993). “The evolution of conventions”. *Econometrica* 61.1, pp. 57–84.

Zenou, Y. and J. Zhou (2022). “Network games made simple”. *Available at SSRN 4225140.*

Zenou, Y. and J. Zhou (2024). “Games on multiplex networks”. *Available at SSRN 4772575.*

Zhou, J. and Y. Chen (2015). “Key leaders in social networks”. *Journal of Economic Theory* 157, pp. 212–235.